Lab 7: Fuzzing for Input Validation Bugs

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Step 2: Buggy application output

```
PS C:\Users\dhriti\Downloads\PES1UG24AM085_SE_LAB_Reports\Lab7_PES1UG24AM085_fuzzing> python hotel_booking.py
==== Running nights_stayed ====
Enter check-in and check-out day (e.g. 12-15): 12-15
Nights: 3

==== Running room_floor ====
Enter a room code (e.g. D-305): D-305
Floor: 3

==== Running booking_label ====
Enter guest names as CSV (e.g. ravi, priya):
Traceback (most recent call last):
  File "C:\Users\dhriti\Downloads\PES1UG24AM085_SE_LAB_Reports\Lab7_PES1UG24AM085_fuzzing\hotel_booking.py", line 31, in <module>
    main()
  File "C:\Users\dhriti\Downloads\PES1UG24AM085_SE_LAB_Reports\Lab7_PES1UG24AM085_fuzzing\hotel_booking.py", line 27, in main
    result = booking_label(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\dhriti\Downloads\PES1UG24AM085_SE_LAB_Reports\Lab7_PES1UG24AM085_fuzzing\hotel_booking.py", line 12, in booking_label
    return lead[0].upper() + lead[1:] + " +" + str(len(names) - 1)
           ~~~~^^^
IndexError: string index out of range
```

python hotel_booking.py on the original code. An empty guest-names input crashes booking_label with IndexError: string index out of range.

Step 4: Fuzz tests failing on the buggy code

```
C:\Users\dhriti\Downloads\PES1UG24AM085_SE_LAB_Reports\Lab7_PES1UG24AM085_fuzzing\hotel_booking.py:2: ValueError: not enough values to unpack (expected 2, got 1)
=========================================== short test summary info ============================================
FAILED test_hotel_booking.py::test_nights_stayed_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_hotel_booking.py::test_room_floor_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_hotel_booking.py::test_booking_label_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_hotel_booking.py::test_all_functions_reject_non_strings - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_hotel_booking.py::test_nights_stayed_checkout_before_checkin_is_invalid - AssertionError: assert -1 is None
FAILED test_hotel_booking.py::test_known_edge_cases - ValueError: not enough values to unpack (expected 2, got 1)
6 failed, 3 passed in 1.13s
```

python -m pytest test_hotel_booking.py: 6 failed, 3 passed. Hypothesis found crashes on None, empty strings, missing separators and non-string input.

Step 5: Fixed application output

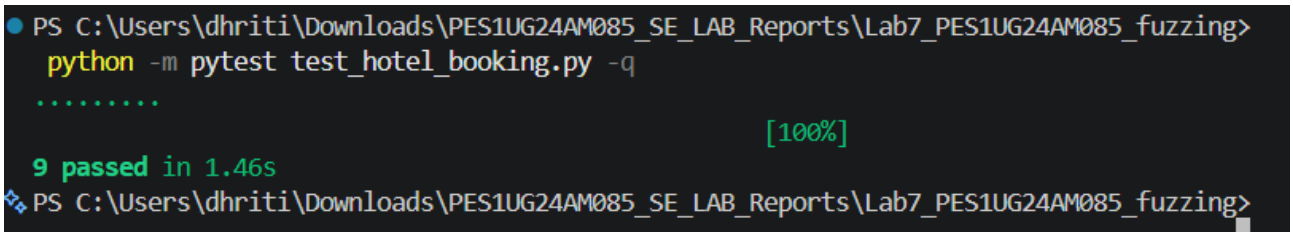
```
PS C:\Users\dhriti\Downloads\PES1UG24AM085_SE_LAB_Reports\Lab7_PES1UG24AM085_fuzzing> python hotel_booking.py
==== Running nights_stayed ====
Enter check-in and check-out day (e.g. 12-15): 12-15
Nights: 3

==== Running room_floor ====
Enter a room code (e.g. D-305): D-305
Floor: 3

==== Running booking_label ====
Enter guest names as CSV (e.g. ravi, priya):
Booking Label: Invalid input
PS C:\Users\dhriti\Downloads\PES1UG24AM085_SE_LAB_Reports\Lab7_PES1UG24AM085_fuzzing>
```

python hotel_booking.py on the fixed code. The same empty input now prints Invalid input instead of crashing.

Step 6: Fuzz tests passing on the fixed code



python -m pytest test_hotel_booking.py: 9 passed.